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10.4 Selection statements

In a program a decision causes a one time jump to a different part of a program. Decisions in C++ are made in several ways, most importantly with if .. else … statement which chooses between two alternatives. Another decision statement , switch creates branches for multiple alternatives sections of code, depending on the value of a single variable.

10.4.1 if statement

The if statement evaluates a condition, if the condition is true then a true-block (a statement or set of statements) is executed, otherwise the true-block is skipped.The general syntax of the if statement is:

```
if (expression)
        true-block;
statement-x;
```

In the above syntax, **if** is a keyword that should contain expression or condition which is enclosed within parentheses. If the expression is true (nonzero) then the true-block is executed and followed by statement-x are also executed, otherwise, the control passes to statement-x. The true-block may consists of a single statement, a compound statement or empty statement. The control flow of if statement and the corresponding flow chart is shown below.

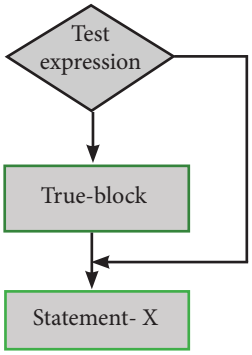

Illustration 10.1 C++ program to calculate total expenses using if statement

```
#include<iostream>
using namespace std;
int main()
{
        int qty, dis=0;
        float rate, tot;
        cout<<"\nEnter the quantity ";
        cin>>qty;
```